

Constructing Modernism: The Cultural Location of Aufbau

1. The Cultural Meaning of *Aufbau*

Between the end of World War I and the immediate post–World War II period, almost a hundred journals and multiauthored volumes appeared in the German-speaking world with the word *Aufbau* in their titles. Practically none existed before the end of World War I, and only a handful remained after 1947. Put into a histogram, the journals fall into three spikes: the largest burst between 1919 and 1927, a middle peak between 1934 and 1937, and a final surge between 1945 and 1947.

These three extraordinary spikes, and the absence of any equivalent spike before 1918 or after the late 1940s, are significant. The first peak corresponds to a staggering proliferation of journals polemically announcing the rise of a new world from the chaos and rubble of the Great War and more specifically from the ignominy and devastation of the German defeat. Out of the rubble, a new structure would emerge; translating *Aufbau* as “rebuilding” or “reconstruction” fails to capture the novelty of what these many authors hoped would come to pass. On new, and for the first time firm, foundations, they would erect a political, philosophical, and aesthetic world separate from everything that had come before. It would (in most instances) be socialist, internationalist, practical, and deeply scientific and technological. As a shorthand designation, I will refer to this cluster of usages as the left-technocratic period of *Aufbau*.

The second peak corresponds to the rise of Nazism, and the *Aufbau* journals and volumes come from both right and left — from the right, organicist volumes espousing work and the empowering of the state; from the left, antifascist journals rising to oppose the destruction already wrought and planned by the dictatorship. Finally, the third *Aufbau* spike was essentially a literature of rebuilding after World War II, a literature of liberal democracy and cultural renovation that would

reintegrate West Germany into the world community; in East Germany, the *Aufbau*-Verlag espoused the building of the new communist society.

What all three moments have in common is this: a rupture with the past, a construction not based on precedent, and a profound conviction that the *Aufbau* process could not be superficial. It had to embody not just the trappings of political change — it had to transform culture, education, architecture, and the modes of reasoning that guide us through the world. Precisely because of its scope, the meaning of *Aufbau* was contested. Right and left both sought to appropriate the impulse to sweep out an old order and to build anew.

This essay focuses on the first spate of *Aufbau* usages. More particularly, I want to exploit *Aufbau* as a keyword (in Raymond Williams's sense [Williams 1983]), a word that embodies a nodal point between culture and society. Even more specifically, the word can help us understand something about the links between cultural domains in the interwar period, for this lexical element was indeed emblematic of a moment at which art, architecture, politics, education, and philosophy could appear to many as participating in a common, quintessentially *modernist* endeavor. Just because the term stood where the tracks of so many lines of culture crossed, groups across the cultural-political spectrum struggled over its appropriation; like modernism itself (for which it is often a stand-in), everyone wanted an *Aufbau* of his or her own design: Nazis, liberals, socialists, and communists.

Though I will make reference to the Nazi and liberal democratic periods of *Aufbau* usage (and to the struggle over its meaning), the spotlight here will be on the left-technocratic period, when the term (principally) connoted the modernist project of building fresh without reference to history, the superficial, or the ornamental. I trace the uses of *Aufbau* in the hope that exploring this particular strand of modernism, located temporally in the interwar years and culturally in the German-speaking lands, will shed some further light on the Vienna Circle's often-repeated self-conception of being at one with the spirit of modernism.

Consider, for example, *Der Aufbau: Flugblätter an Jugend* (*Der Aufbau: Leaflets for youth*), published in 1921, which included an article by Hans Reichenbach on socialism (see fig. 1). The introductory essay in the series pursued the relationship between *Aufbau* and revolution:

What is *Aufbau*? Is *Aufbau* "changing the world"? Evidently not! For what changes the world more than revolution? And yet we say: a revolution creates only the conditions for an *Aufbau*. Change is attached to the given, to the substantial; it is a redirection, a shifting of forces;

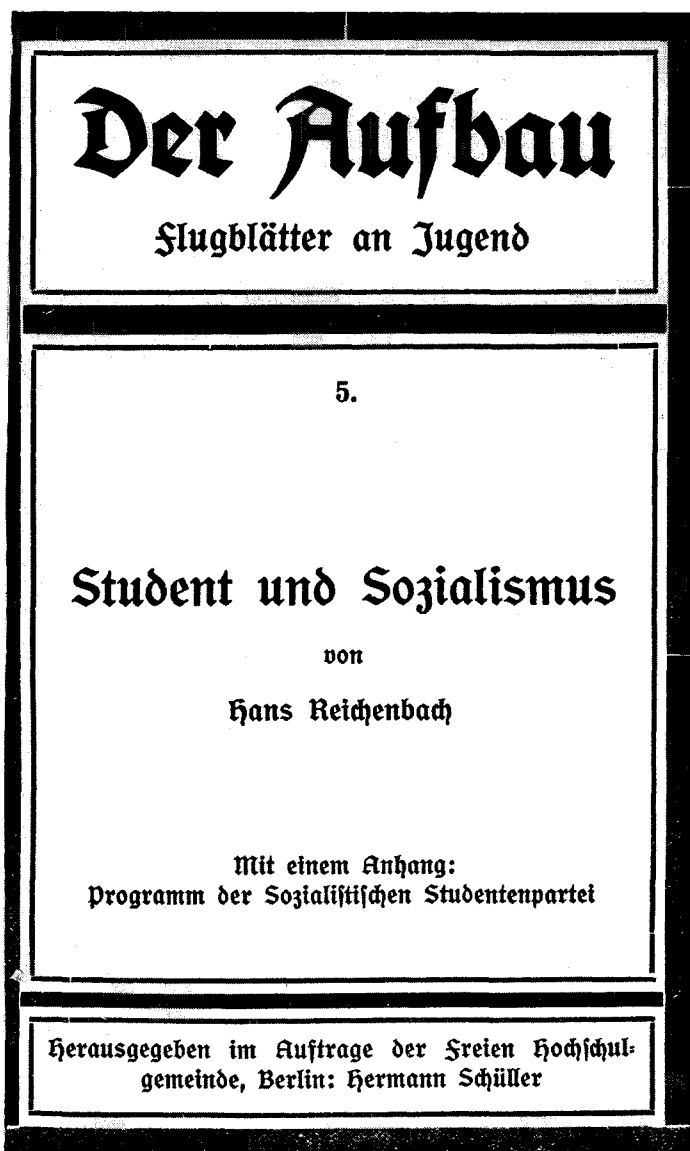


Figure 1. Cover page from volume 5 of *Der Aufbau: Flugblätter an Jugend* (1919), featuring work by Hans Reichenbach. From Harvard University, Widener Library, Cambridge, Massachusetts.

it is the technical. (Schüller 1921, 1:5; see also Reichenbach 1919, 5:3–9)

Even this brief excerpt captures themes that draw together many of these manifestos. For in this postwar time of revolutionary governments, from Russia to Bavaria, there was a sense that politics alone failed to plumb the depth of the transformation necessary in society. Means of transport, modalities of production and distribution, the architecture of buildings, and the planning of cities — these material and technical realms held out a real and symbolic importance that for many lay deeper than politics alone. At the same time, in many of the *Aufbau* journals, this emphasis on the need for change at the deepest level, below politics, so to speak, was coupled with an almost mandatory reference to the destructiveness of war that had made it necessary. This was explicitly the case in the journal *Umsturz und Aufbau* (Destruction and construction), which began publishing in 1919, as the following excerpt indicates:

We have lived through the most unheard of catastrophe that a people has ever encountered. In need and death, in blood and tears there has been only one asset — but a powerful one — left to us: the spirit. To make the spirit fruitful for the renewal of our people is the goal of our series *Umsturz und Aufbau*. Here we want to assemble documents. . . . [W]e want to remove the rubble of the European killing fields and with heart and head to prepare for the erection of a new humanity. (*Umsturz und Aufbau* 1919 [1974], 22–23)

Digging out from the catastrophe was a trope repeated over and again, as Europeans — especially in the German-speaking world — contemplated their present and their future. The left-leaning Swiss periodical *Der Aufbau: Sozialistische Wochenzeitung* concurred that the war years had shed light on the “catastrophe” but saw in socialism an exit from the destruction (see fig. 2). In the first edition (December 1919) of the journal, the editors pronounced that their weekly publication would show itself to be

born out of the need of our time, . . . aimed at those who, like us, feel that need. . . . In these war years, they have experienced the last horrible but inevitable consequences of one of the innermost laws of life of a fallen world. They know that if they do not want to be destroyed, then a new existence [*ein Neues*] is necessary. A renewal, with its roots in the deepest essence of humanity, but which for that reason also is expressed in the outer structures of life, must transform them. *Der Aufbau* must proceed on new fundamentals and according to new principles. (*Der Aufbau: Sozialistische Wochenzeitung* 1919, 1)

DER AUFBAU

SOZIALISTISCHE WOCHENZEITUNG

Erscheint jeden Freitag

No. 1

Zürich, 5. Dezember 1919

Abonnementpreise:
Jährlich Fr. 6.50; 1/2-jährlich Fr. 3.50; 1/4-jährlich Fr. 2.—. Ausland: Jeweiliger Portozuschlag.
Preis der Einzelnummer 20 Cts.

Herausgeber: Genossenschaft zur Herausgabe einer sozialistischen Zeitung.
Redaktion: MAX GERWIG, MAX GERBER.
Druck, Administration und Expedition: Pflingtwaldstrasse 57, Zürich 5.

Insertionspreise:
Die einspaltige Feuilleton 30 Cts. Ausländische Anzeigen 50 % Zuschlag.

Was wir wollen.

I.
Was dieses neue Wochenblatt will, wird es selbst zeigen müssen. Geboren aus der Not unserer Zeit, wendet es sich an alle, die mit uns diese Not empfinden. Wir wissen, dass ihrer auch in unseren Völkern nicht wenige sind. Sie haben in den vergangenen Kriegsjahren die letzten schauerlichen, aber naturnotwendigen Konsequenzen einer von den innersten Lebensgesetzen abgefallenen Welt miterlebt und wissen, dass, wenn sie nicht zugrunde gehen wollen, ein Neues nötig ist. Eine Erneuerung, die ihre Wurzeln freilich im tiefsten Wesen des Menschen selbst haben, aber gerade deshalb auch in den küssen Lebensordnungen zum Ausdruck kommen, sie umwandeln muss. Der Aufbau muss auf neuen Fundamenten und nach neuen Prinzipien erfolgen. Noch nie wohl war deshalb die Zeit dem sozialistischen Ideal so günstig als jetzt, wo die bisherige Wirtschaftsordnung im Lichte katastrophaler Ereignisse und im Lichte ihrer unzulässigen Fragen gerichtet erscheint. Aber wie jedem Ideal, so drohen nun auch dem Sozialismus desto grössere Gefahren, je näher er seiner Verwirklichung reift. Da ist einmal die Bedrohung von aussen. Seit in der russischen Revolution das sozialistische Ideal greifbare Wirklichkeit anzunehmen schien, seit es klar wird, dass der Kampf nicht um ein paar Reformen mehr oder weniger, sondern um eine grundsätzliche Neugestaltung der Wirtschaft geht, haben sich alle Mächte der Reaktion zu einem entscheidenden Verweigerungskampf aufgeführt und suchen mit allen Mitteln die Entwicklung zum Sozialismus zu hindern. Und doch liegt hier die kleinere Gefahr, denn der Sozialismus kann nicht von aussen, durch die absterbenden Mächte einer vergangenen Epoche überwandelt werden. Der Sieg in diesem Kampfe ist ihm gewiss, wenn er nur sich selbst, seinem inneren Wesen treu bleibt. Aber die ungeheure Gefahr ist, dass Schlandrian und Nachahmung sich auch der Revolutionäre bemächtigen und sie zu Phantomen des Radikalismus des lösenden Wortes und der Gewaltgebräde machen, dass sie nicht wissen, und nicht wissen wollen: die Umwandlung der Gesellschaft kann nur in Liebe, in Arbeit, in Stille kommen. (Gustav Landauer.)

Wir begreifen nun freilich, wie darauf, dass die in Not und Elend versunkenen Massen der bolschewistischen Gedanken, durch Gewalt und vorübergehende Diktatur

in kurzer Zeit das Alte zu stürzen und eine neue Ordnung sozialer Gerechtigkeit und Freiheit zu errichten, wirken muss. Wir wissen, wie viel Idealismus und verführerischer Glaube in den Zielen des Bolschewismus steckt. Wir wissen auch, dass auch seine Methoden nur zum kleinsten Teil aus niedrigen Gewaltinstinkten und Leidenschaften, vielmehr aus grosser Hoffnung und grosser Entschlossenheit geboren sind. Die grossen Massen können den Gedanken nicht ertragen, dass das gewaltige Völkerringen mit seinen Verwundungen zu Ende gehen soll, ohne dass der Grund gelegt werde zu einer andern, solidarischen und brüderlichen Welt. Sie haben, zu oft schon getäuscht und betrogen, kein Vertrauen zu den Versprechungen bürgerlicher Machthaber und Parlamente, und sehen nur noch den Ausweg gewaltsamer Revolution. Schliesslich betonen wir, dass gerade die Vertreter des Bestehenden am wenigsten Grund haben, sich über den Bolschewismus zu entrüsten, denn der in ihm steckende Gewaltgeist ist der Geist der bestehenden Ordnungen und Mächte; der Bolschewismus will nichts als die Probleme mit denselben, vielleicht etwas vergröberten Methoden lösen, die ihm Geschichte und Gegenwartspolitik gelehrt haben. Aber gerade hierin liegt nun auch sein Genetium und sein Verhängnis. Denn es ist eine tiefe, mit der Sicherheit eines Naturgesetzes wirkende Wahrheit, dass das Gute seine eigenen Kampfmethoden hat und dass eine Sache nur durch diejenigen Mittel wirklich werden kann, die ihrem innersten Wesen entsprechen. Ein Reich der Gewalt kann mit Gewalt errichtet werden, die Freiheit kann nur realisieren, wer an die Freiheit glaubt und es mit ihr wagen will. Wo der Sozialismus mit ihm wesensfremden Methoden zum Ziel gelangen will, da gibt er etwas von seinem innersten Wesen, seiner dem Bestehenden überlegenen Kraft preis und verlässt dadurch sich selbst. Gerade das, was gerade von den Mächten der Gegenwart, dass auch diejenigen, die etwas Neues möchten, in so hohem Grade selbst dem „Zeitgeist“ zum Opfer gefallen sind, dass auch das sozialistische Ideal vielfach verflüchtigt wird gerade von den Mächten aus, die überwinden möchte; dass wir auch in der Sozialdemokratie, in deren politischen Zielen doch eine neue Wahrheit aufleuchtet, so oft dieselbe geistige Einstellung finden, die das Verderbliche unserer Kultur ausmacht: den Glauben an das Quantum, an die Zahl (z. B. die Mitgliederzahl), den Glauben an die Gewalt, den Aberglauben an das Wissen (z. B. in der Bildungsarbeit), die Unberechnung der Wissenschaft, die un-

sere heutige Wirtschaftsordnung tragenden Gedanken, dass nur der Appell an das Interesse, an den Egoismus den sozialen Bau aufrechterhalte usw. Und doch sind wir fest überzeugt, dass durch nichts jetzt so sehr der Geist der alten Ordnung und damit diese selbst gestärkt werden könnte und das Leben der heutigen Gesellschaften und Wirtschaftsformen künstlich verankert als durch einen Sozialismus, der stecken bleibt im bürgerlichen Glauben an die Gewalt, in der lähmenden Skepsis gegenüber dem Geist und den idealen Triebkräften; der Gewaltgeist würde sich nur allzu bald wieder gegen den Sozialismus selbst kehren und ihm den Todesstoss versetzen, denn er ist von Natur aus der beste Verbündete derer, denen Gewaltherrschaft und Unterdrückung nicht nur Übererzangeepochen sondern auch Endziel sind. Jedenfalls ist ein solcher Ausgang sehr wahrscheinlich als die aller historischen und psychologischen Erfahrung widersprechende Utopie, man könne die bösen Geister, die man gerufen, nach volbrachtem Dienst nach Belieben wieder beurlauben und so auf dem Wege vorübergehender Despotie in das Reich der Freiheit gelangen.

Es liegt aber im Wunderglauben, dass Alle auf einen Schlag stürzen und auf neuen Trümmern das Neue erbauen zu können, noch eine weitere Gefahr. Er lenkt ab von den drängenden und bitter notwendigen Vorbereitungsaufgaben und den gegenwärtig schon vorhandenen Möglichkeiten. Das Interesse für sie erlahmt, weil sie im Lichte einer augenlich nahen sozialistischen Zukunft überflüssig scheinen. Die neue Wirtschaft ist aber kein Mechanismus, den man nach Eroberung der Macht konkretisieren könnte wie eine neue Maschine, sondern ein Organismus, der aus tiefen Wurzeln wachsen und reifen muss. Wer u. a. im Hinblick auf das russische Beispiel die ungeheure Schwere der Aufgabe empfindet, nach dem Zusammenbruch des Kapitalismus die bisher im Prinzip durch den Anreiz des Profites getriebene Wirtschaft in neuen Formen und mit ganz anderen Triebkräften aufrecht zu erhalten; wer weiss, dass der Kapitalismus nicht das Wunder tun wird, in seinem platzenden Zusammenbruch den Sozialismus zu gebären, der wird schon jetzt nach den neuen wirtschaftlichen Formen suchen und sie nach Möglichkeit verwirklichen, um so wenigstens das Bett zu graben, in das sich einmal der Strom neuen Lebens ergiessen kann. Es gilt, mitten im bestehenden Wirtschaftssystem von unten her nach neuen Prinzipien aufzubauen. In der Rede, dass

Figure 2. Title page from the first issue of the Zurich-based Der Aufbau: Sozialistische Wochenzeitung (December 1919). From Harvard University, Andover-Harvard Theological Library, Cambridge, Massachusetts.

Socialism was this transforming force and contributed the principles upon which these inner and outer mutations would build. As in so many of the journals, the authors of this one went on to elaborate their vision of a mutation in society that would not end with the “merely” political:

We do not want to stop at politics in the narrow sense, for the new reality that we hope for is not simply a new political or economic system, but a renewed world from the depths on up. [This renewal] must come to a breakthrough in all other domains of life: public education, comradeship, the essence of child rearing, science, art, and religion. (ibid., 1)

Although in the period 1919–26 the *Aufbau* journals and other volumes tended to be left-technocratic, others already explored aspects of nationalism and Groß-Deutschland. From what we know of the contested history of modernism in the Nazi period (see, for example, Herf 1984), it comes as no surprise that *Aufbau* too was seized upon by the national socialists. *Aufbau und Erbe*, a multiauthored volume, was just one of the *Aufbau* texts sporting a swastika on its cover. Its content, ranging over music, poetry, and prose, returned to the Nazi themes of land, weapons, and handwork; *Aufbau* in this context meant building up the new fascist society, including the new fascist man. Indeed, the scope of the fascists’ project of building up is no less thoroughgoing than that of the left. Verses tell of building weapons, of defending the fatherland, of physical strength, and of work. It is an *Aufbau* altogether different from the transparent construction that characterized the Dessau Bauhaus or the Vienna Circle.¹

Precisely to oppose everything the fascist buildup envisioned, especially after the Nazi *Machtergreifung* (seizure of power), a spate of anti-Nazi *Aufbaus* emerged. One, founded on 1 December 1934 in New York, grew into a highly influential journal published in German primarily by and for German Jews. They announced in their *Aufbau Almanac*:

Its name [*Aufbau*] is its program. Over there, in Europe, every pinnacle of achievement that the German-speaking Jewish people have erected has been leveled. In a few years, centuries were annulled. In the monstrous waves of hate that flooded a flourishing cultural landscape, that which in a thousand ways had organically grown and often had matured to the most beautiful bloom was scattered to the winds. . . . Resurrection on a new Earth, on new roots, the focusing of all creative elements to this purpose — that is the first task that the *Aufbau* serves. (*Aufbau Almanac* 1941, 6)

Between 1937 and 1945 the number of new *Aufbau* journals declined. But following the end of World War II they reemerged in large numbers, founded by German speakers flung from Österreich to Ohio. In Vienna in July 1946, the *Stadtbauamt* (office of city planning) began a monthly *Aufbau* dedicated to *Wiederaufbau*. Bundespräsident Karl Renner penned a dedication lamenting the terrible destruction wrought by the modern day "Caesars" and extolling the Viennese to follow the special course of Austrian art and architecture in the rebuilding of the city. Others reiterated the necessity of rebuilding but at the same time mourned the destruction that had been deeper than merely material, calling for a reconstruction that must be so as well:

The old world is destroyed.

Before its ruins stand an old and a young generation, both suffering greatly from the unhealthy consequences of a false politics, a poorly applied technology, and a failed organization.

Now we, the people, must take a new path. Each person must find his own spiritual way, but in everyday life, in the practical, we must go together. . . .

Reconstruction is preponderantly a technical problem. The organizer, the engineer, the architect will carry it out. . . .

Der Aufbau is oriented towards all technical people, but beyond that to everyone struggling for the new form of society. (Körner 1946, 2)

Even in the far-flung POW camps in the United States, German prisoners gathered to create a journal entitled *Der Aufbau: Lagerzeitung für Camp Perry und Nebenlager* (Der Aufbau: Prison newspaper for Camp Perry and neighboring prison camps). The typescript paper, with its symbol of a rising sun, announced in one of its first numbers that the history of Weimar had to be rewritten not as one of destructive chaos, but as the attempt at a new beginning. Linking that postwar time with their own (October 1945), the prisoners looked to a future that would, once again, build democracy out of the rubble (*Der Aufbau: Lagerzeitung für Camp Perry* 1945).

This is, of course, just a sampling of the literature of *Aufbau*. But with this plethora of *Aufbaus* in mind, we can begin to piece together what the cultural resonance of the term would have been in the post-World War I years. At the first, most direct level of interpretation, the term suggested a quite literal construction. Architecture played a crucial role because of the great need for housing necessitated by the massive urbanization that followed the Great War. In Vienna alone, between 1922 and 1925 some 20,600 dwelling units were added to the city's stock (Schus-

ter 1926, 8). Second, as is well known, the new buildings, especially the *Siedlungen* (mass housing), became a principal focus of the modernist architecture in both Austria and Germany.² Political and architectural progressives clustered around this chance to shape the daily lives of workers. Finally, in each of the periods, left-technocratic (1918–26), Nazi/anti-Nazi (1933–37), and liberal-democratic (1945–48), the notion of *Aufbau* was taken far beyond the exigencies of mortar, steel, and glass. From the merely material, the notion of construction was elevated to a program for a way of life.

Otto Neurath bridged the two conceptions of *Aufbau*: the material building up and the construction of an ordered, rational *Lebensform*. Both, Neurath contended, were of the utmost importance following the terrible destruction of the Great War. On one side, he reacted to this devastation like so many others, with horror. Yet curiously he found a constructive aspect to the war economy; he saw it function with extraordinary results, drawing resources from a centralized agency and deploying them to maximum effect. In April 1919, Neurath, then president of the Central Office of Economics for the Munich Workers' Council, had responsibility for the socialization of Bavaria. Having passed from critic to actor, he now set about enforcing the peacetime analogue of wartime centralization (Neurath 1973, 124).³

With the war over, part of the challenge of centralization and planning was the immense problem of coping with the huge migration of Austrians from the countryside to Vienna. Neurath followed the new movements in housing with great care.⁴ He organized propaganda for the *Siedlungen*; he attended lectures by contemporary architects such as Walter Gropius; and he worked in close collaboration with one of Austria's leading architects and architectural theorists, Josef Frank (Philipp Frank's brother). Collaborating on a range of projects, from museums to the *Isotype* system of universal signage, the sociologist and the architect came to share a vision of modernity that joined aspects of reformist Viennese architecture to elements of German architecture with its "scientific" approach to a myriad of social problems. Neurath and Frank were sufficiently in sympathy that Josef Frank was invited to give the first in a series of lectures to the Verein Ernst Mach in 1929. Not long afterward, Neurath, Carnap, Philipp Frank, and Hans Reichenbach all lectured at the Dessau Bauhaus, then the citadel of modernist architecture.⁵

Hannes Meyer, the Swiss architect, theorist, and Marxist who succeeded Gropius as director of the Dessau Bauhaus in 1924, invited all of these philosophers to give the lectures. Meyer's celebration of the technical was boundless. In one of his most famous articles, "The

New World,” he began with a staccato hymn to the machine world: “Flash! The “*Norge*” has circled the North Pole! Completed! The Zeiss Planetarium at Jena. To be tested today: Flettner’s rotorship! These were, in 1926, the most recent exploits in the mechanization of our globe” (Meyer 1926). Airplanes, ships, parachutes, gliders, microphones, gramophones — these and many other devices were (by Hannes Meyer’s lights) changing the nature of space and time, and at the same time destroying patriotism and inculcating internationalism. Art as well as politics yielded to the new complex of gears and lenses: “The artist’s studio becomes a scientific-technical laboratory, and his works are the fruits of analysis and invention” (ibid.). Neurath had nothing but appreciation for Meyer precisely because Meyer (unlike Gropius) had explicitly put engineering front and center. But the technical in Meyer (as in Neurath) was more than the physical instantiation of instrumental reason: the technical stood for and created a way of life, an orientation toward the world that would cleanse it of the full spectrum of irrational threats: above all clericalism, nationalism, and social divisiveness (see Galison 1990, 709–52).

At the time Meyer wrote his “New World,” Neurath himself was deeply engaged with the direction and writing of a journal entitled (not surprisingly) *Der Aufbau* (1926) (see fig. 3). Sharing the masthead with him were the Austrian architects Franz Schuster and Franz Schacherl. Schuster, collaborating with Adolf Loos, Schacherl, and others, had participated in the design of some 1,589 apartments by 1926. Heinrich Tessenow, also on the directorship of the journal, was the head of the architecture school of the Dresden Academy of Art and had been professor in the Vienna *Kunstgewerbeschule* since 1919. Like Schuster and Schacherl, Tessenow too had participated in the Red Vienna architecture, contributing (for example) to the Kolonie Rannersdorf outside Vienna.⁶ Martin Wagner, the Berlin architect, was *Stadtbaurat*. But probably the best known of the journal’s leaders was another Berlin architect, Bruno Taut, creator of the famous Glass Pavilion for the Cologne Werkbund exhibition of 1914. In 1919, Taut and Gropius, two leading members of the Berlin *Arbeitsrat für Kunst* (Working council for arts), had drawn up a program similar to the one Gropius was setting out at just this time for the Bauhaus.⁷

The Viennese *Der Aufbau* specified its orientation in its subtitle, *Österreichische Monatshefte für Siedlung und Städtebau* (Austrian monthly for mass housing and city building). But its pages contained far more than plans for walls of reinforced concrete and comfortable rooms for the working class. Their founders’ hope, expressed over and again in philosophical, methodological, architectural, and sociological tracts,

DER AUFBAU

ÖSTERREICHISCHE MONATSHEFTE FÜR SIEDLUNG UND STÄDTEBAU

HERAUSGEBER: ARCHITEKTEN FRANZ SCHUSTER - FRANZ SCHACHERL

MITARBEITER: REGIERUNGSRAT A. D. DR. HANS KAMPPFMEYER (WIEN),
GENERALDIREKTOR DR. ING. HERMANN NEUBACHER (WIEN), DR. OTTO NEURATH (WIEN),
STADTBÄURAT A. D. ARCH. BRUNO TAUT (BERLIN),
PROFESSOR DR. HEINRICH TESSENOW (DRESDEN),
STADTBÄURAT A. D. DR. ING. MARTIN WAGNER (BERLIN).

NUMMER 1

WIEN, IM FEBRUAR 1926

JAHRGANG I



UNSER WOLLEN.

Wir wollen mitwirken am kulturellen, sozialen und wirtschaftlichen Aufbau der Gesellschaft, die aus der gegenwärtigen Unklarheit, Unordnung und chaotischen Verworfenheit, nach Klärung und Ordnung strebt.

Unsere Zeit fordert von jedem Einzelnen und von der Gemeinschaft äußerste Anspannung und vornehmste Besinnung, um einer neuen Entwicklung eine richtige und bedeutende Bahn zu geben.

Jede Zeit hatte ihre Verantwortung vor der Zukunft, aber keine seit langem hatte sie so sehr wie unsere. In rascher und fast beängstigender Aufeinanderfolge hat das 19. Jahrhundert Erfindungen und Entdeckungen gezeitigt, die Jahrhunderte alte Träume und Märchen in besonderer Weise wahr gemacht haben. Geniale Maschinen und unheimliche Instrumente stehen uns zur Verfügung, um die weitverzweigten Bedürfnisse des Lebens in einer Art und Weise zu erfüllen, wie dies keine Zeit vor uns gekannt hat; den großen wissenschaftlichen und technischen Umwälzungen folgten politische und soziale. Alles steht unvermittelt neben Neuem, Chaos, Unklarheit, Unordnung, ist unser Erbe nach dem vergangenen Jahrhundert und unsere Städte, unsere Häuser und unsere Wohnungen sind ein getreues Abbild dieser Zerrissenheit und Haltungslosigkeit! Unser Jahrhundert scheint berufen, wieder zu ordnen und zu scheiden, dem Alten seinen ihm zukommenden Platz zu geben und dem Neuem sein Recht!

Es soll unsere Aufgabe sein jene Probleme zu erörtern, die im Bauen auf Umwandlungen, Umstellungen und neue Wege hinführen.

„Bauen“ ist für uns nicht eine Angelegenheit des Einzelnen, des Bauherrn, des Architekten oder des Baumeisters, sondern der Gesamtheit.

Wir verurteilen daher die Einstellung, daß jeder Einzelne machen könne was ihm behage und wie er es nach Laune und persönlichem Dafürhalten wünsche; wir wissen, daß dies im Gegensatz steht zur individualistischen Auffassung breiter Schichten, aber wir wissen auch, daß nur größte Rücksicht auf die Gesamtheit, daß nur ein ernstes Eingehen auf die Forderungen und Notwendigkeiten eines großen Gesamtwillens, den häßlichen und unhaltbaren Zustand unserer Städte überwinden hilft, der allen am Aufbau einer neuen Gesellschaft Arbeitenden in kultureller, sozialer und wirtschaftlicher Hinsicht die ernstesten Sorgen bereitet.

Der zerrissene, würdlose Großstadtrand mit seinen verwahrlosten Zinshausblöcken und Feuermauern dicht neben dem Ackerland und der alten Dorfstraße, ist uns ein Zeichen der Zerrissenheit und Würdelosigkeit unserer Zeit, deren Überwindung ein Ziel aller sein muß. Wir sind überzeugt, daß dies nicht geht ohne die breiten Massen, ohne daß die ganze Bevölkerung an diesen Fragen verstehenden Anteil nimmt, Aufklärung zu geben und den Kampf zu führen, soweit es im Rahmen und der Möglichkeit einer Zeitschrift liegt, soll Aufgabe des Aufbaus sein.

Wir betrachten es als schwächlich und verlogen, den heutigen Zustand etwa mit einem Hinweis auf die „gute alte Zeit“ zu einer freundlichen Veränderung bewegen zu wollen; wir glauben nicht an die Idylle der Postkutsche und die „Schönheit der gotischen Stadt“, weil nicht umsonst die schöpferischen Köpfe der Menschheit aus der Postkutsche die Eisenbahn und das Auto und das Luftschiff gemacht haben und weil wir wissen, daß zum Beispiel das gotische Wien in allen Straßen so voll Schmutz war, daß jeder Bauer, der in die Stadt kam, eine Fuhr Unrat hinausführen mußte!

Figure 3. Title page of the first issue of *Der Aufbau: Österreichische Monatshefte für Siedlung und Städtebau*, 1/1 (Vienna, February 1926).

was to create a new *Lebensgestaltung*, a new form of life. Housing pressure in Vienna began mounting even before World War I. Peasants were moving to the city from across the empire; roughly fourteen thousand new units were going up every year, and even that number could hardly keep pace with demand. During the war itself, construction essentially ceased (372 apartments in 1917), leaving a four-year gap in building. Following that lag in construction came a massive postwar influx of soldiers, the burst of new families, and droves of people displaced by the end of the Hapsburg regime. It is not hard, then, to see why politicians of every stripe put the housing situation front and center. Land on the outskirts of Vienna filled first with vegetable patches, then with animals and barns, and eventually with shanty-town dwellings. By 1920 some thirty thousand people who populated these “wild settlements” began organizing and launched two massive demonstrations that stunned Vienna. The newly elected socialists, under Otto Bauer, responded with donations of plumbing, windows, and doors — soon factories went up to provide these items in ever-increasing numbers. Appropriation of “underutilized” apartments began in earnest as did the seizure of unused lots, factories, and commercial space. Within a few years, the government distributed some ten thousand such units.⁸

But the most dramatic development was the beginning, in 1921, of the *Wohnbausteuer*, a highly progressive rent tax earmarked for the building of radical new *Siedlungen* — giant complexes of workers’ apartments surrounded by stores and including kindergartens, laundries, and a collectively owned courtyard. These would be, by socialists’ parlance, the “palaces” of the proletariat. From the beginning, these buildings, such as the Karl Marx Hof, were intended not merely to house but to transform the working class: to bring light and air into houses, to banish forever the backyard dark shacks that demeaned the workers. The new apartments would be space and work efficient, and so foster collective values of property; the new architecture would free women from domestic labor and instantiate the *factual* rationality that was the workers’ birthright. This was a self-consciously politicized architecture that would exploit space, material, and design to combat the older bourgeois architecture. Older buildings divided families from each other and led people to cling to a dying past. Architects of Red Vienna thought of themselves as building — literally and figuratively — a new life (Marcuse 1986, 558–85). Perhaps as important as workers, the *Siedlungen* became the primary exhibition point for the host of new architects who were emerging from Austrian and Bauhaus movements. What was needed was an *Aufbau*.

The polemical opening of this particular *Aufbau* (the one that Neu-

rath, Schuster, Schacherl, Wagner, Taut, and Tesserow directed) de-claimed:

We want to work together on the cultural, social, and economic *Aufbau* of society, which is striving from the present unclarity, disorder, and chaotic confusion toward clarity and order.

Not only in the manifesto, but in all subsequent issues, the journal made it plain that it despised the ornate in architecture; elimination of the “stylish” was essential in all aspects of life. Reforming the “interior” meant rationalizing personal life as well as public buildings. Together, this restructuring of the personal and the public would mesh people like precision gears into the technical-objective world. Modernity, in the form of this constructed world, opposed the romantic, the religious, the national, and the traditional. The *Aufbau* manifesto continued:

We take it as weak and untruthful to wish to move the present situation to a “friendly” change with reference to the “good old times.” We do not believe in the idyll of the postal carriage and the “beauty of the gothic state” because not for nothing did the creative minds of mankind make the train, the car, and the airship out of the postal coach; and because we know, for example, that gothic Vienna was so full of dirt in every street that each farmer who came to the city had to take out a cartful of filth! (*Der Aufbau: Österreichische Monatshefte* 1926, 1/1:1)

Modernity, for Neurath and his colleagues, was precisely the tearing down of this gothic Vienna and its replacement by the rational and factual (*sachlich*). In his first individual contribution to the journal in May 1926, “Rationalismus, Arbeiterschaft, und Baugestaltung,” Neurath began by berating the lip service paid to modern architecture:

The demand “construction projects should be handled objectively [*sachlich*]” is almost acknowledged by the architects of late capitalism. The demand of an architecture free from ornaments and decorations, which views buildings like a kind of machine, is self-evident. Nevertheless, it occurs so seldom despite the fact that it is spoken about so often.

Deriding the ornate façades of the Palace of Justice and the War Ministry in Vienna, Neurath contrasted approvingly the plan and appearance of a factory designed in part by Gropius as exemplary of the new rationalism (Neurath 1926b, 49).

Central to Neurath’s analysis was his belief that an inexorable march of technology had transformed society, extending from social hygiene

to the inner workings of the mass production factory. All those who held out against these developments (the train tracks, telephones, and telegraphs) were standing in "crass contradiction" to the rationalization that could be seen everywhere. The educated and upper classes — even artists and scientists who should know better — still resisted the physical instantiation of rationality into their homes. Nothing illustrated this recalcitrant attachment to traditional ways of life better, Neurath argued, than the ornate bronze lamps that took the style of gas lamps at a time when electrical bulbs had made their form utterly inappropriate (see fig. 4). This reluctance to shed the forms of a previous and now-outmoded life was particularly acute in Austria, Neurath contended, because the bourgeoisie lacked the factual (*sachlich*) tradition. In England, by contrast, the bourgeoisie had the independent existence needed to match inner and outer forms (ibid., 51).

All these vestiges of the past went together: the romantic way of thinking in private life, nationalism, clericalism, traditional furniture and clothing. Just because Neurath saw this full circle of cultural and political goals as intertwined into a *Lebensordnung* (order of life), *Lebensgestaltung* (form of life), or *Gesamtgestaltung* (total formation), he interwove political advocacy of a classless socialist society with a critical commentary on the constructivist paintings of Fernand Léger. Such art exemplified the superficial adherence to the hard-edged rationalism of modernism. To Neurath, the artist who appropriated the symbols of modern means of production without grasping production methods themselves resembled nothing so much as a child playing with retorts and test tubes without understanding the underlying chemical reactions. The reproduction of bits and pieces of machines was, in and of itself, meaningless. Outer forms counted for nothing. Architects would do far better to show the public blueprints than to show them façades. Is a house light and airy? That matters; stone cornices do not (ibid., 52). Real construction, an *Aufbau* based on simple structural forms and on rational principles, was needed.

For of all the elements of private life, it was in architecture and factory production that the rationality of the working class would be expressed in its objective-technical form. Rationalization penetrated everyday life not by the visible gears of machines, but by the thoroughgoing saturation of workers' lives through Taylorism in the giant factories of the late nineteenth and early twentieth centuries. Still, by Neurath's lights, it was by no means a foregone conclusion that the forces of *Sachlichkeit* would succeed, for arrayed against them was an Austrian "ruling class" that, far more than its British equivalent, was hostile to the new, more rational way of life (ibid., 51). To Neurath,

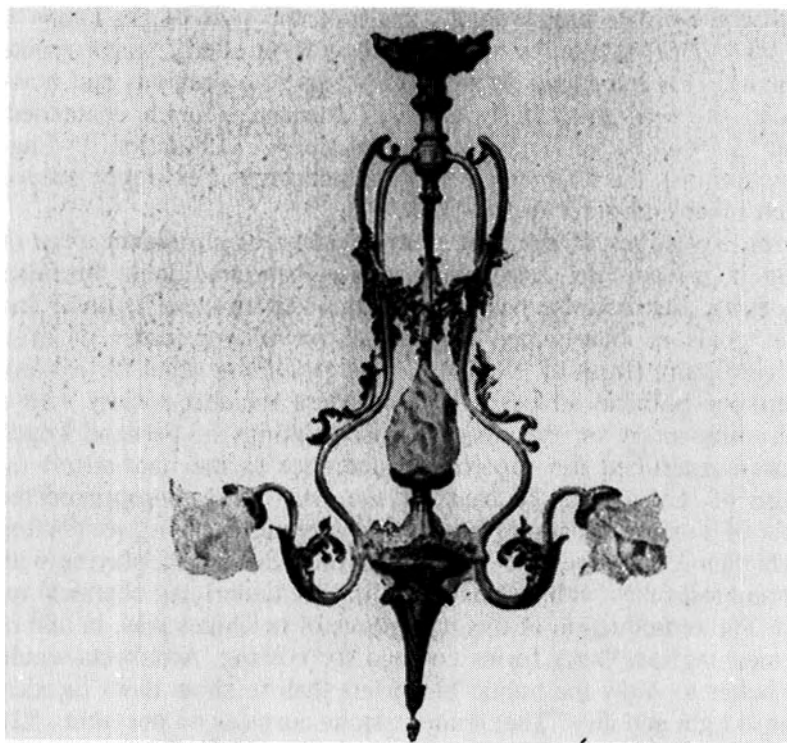


Figure 4. Electric lamp with the appearance of a gas lamp. From Der Aufbau: Österreichische Monatshefte für Siedlung und Städtebau, 1/1 (Vienna, February 1926).

only the working class lay free of the claims of the past; only the working class saw the *Lebensordnung* of everyday life as a means to its happiness:

To such a way of thinking [the proletariat's] architecture is a slice from the total form of life [*Gesamtgestaltung des Lebens*]! Its task is to grasp clearly this goal and, in its realm, to realize it. In architecture the workers' rationalism finds its new expression: in its factual-technical forms [*sachlich-technischen Formen*]. (ibid., 54)

Neurath and his colleagues on the board of *Der Aufbau* celebrated their ostensive break with the past. In one issue they contrasted a full-page collage of ancient monuments and statuary with icons of electricity, shipping, power, and factories (see fig. 5). Too many Sundays had been lost in the accumulation of antiquities, the first page proclaimed. Now the new world was at hand, and humankind's duty was to be worthy of it. "Art," classically conceived, was to be replaced by all that was rational and therefore international. The watchwords were *Welthandel*, *Weltindustrie*, *Weltverkehr*, and *Weltwirtschaft* (*Der Aufbau: Österreichische Monatshefte* 1926, 1/6:99). World structure and inner life were bound together: modifying even the vocabulary of expression became a way of modifying thinking.

Neurath's own writings on the necessity of bringing interior and exterior life into line with the time were thus part and parcel of a journal project within the larger literature and politics of post-World War I *Aufbau*. And it was within this particular circle of *Aufbau* connotations that Neurath's Vienna Circle pronouncements on "inner and outer life," and on an embattled but inevitable *Sachlichkeit*, must be understood.⁹

Perhaps most significantly, Neurath identified contemporary architecture with the creation of new ways of thinking, acting, and feeling, and he wrote article after article about substituting the hard-edged, hard-boiled rationalism of technical objectivity for false fronts and façades. The reformation (or perhaps creation *de novo* of the "modern man") was explicitly tied to the establishment of new "tools for living" associated with an age of ordered, structured reasoning and mechanical operations.

In December 1926, the year Neurath and his collaborators founded their *Aufbau* journal, the startling edifice of the Dessau Bauhaus opened. Perhaps inevitably, Neurath was there. The building's sharp-edged lack of ornamentation pleased him, though he reiterated his (Josef Frank-like) caution against elevating functionalism into a new stylistic fetish of its own. "When will the modern engineers run the Bauhaus?" he asked. For if the Bauhaus was truly to usher in the age of a new form of societal and personal life (*Neugestaltung des gesellschaftlichen und*

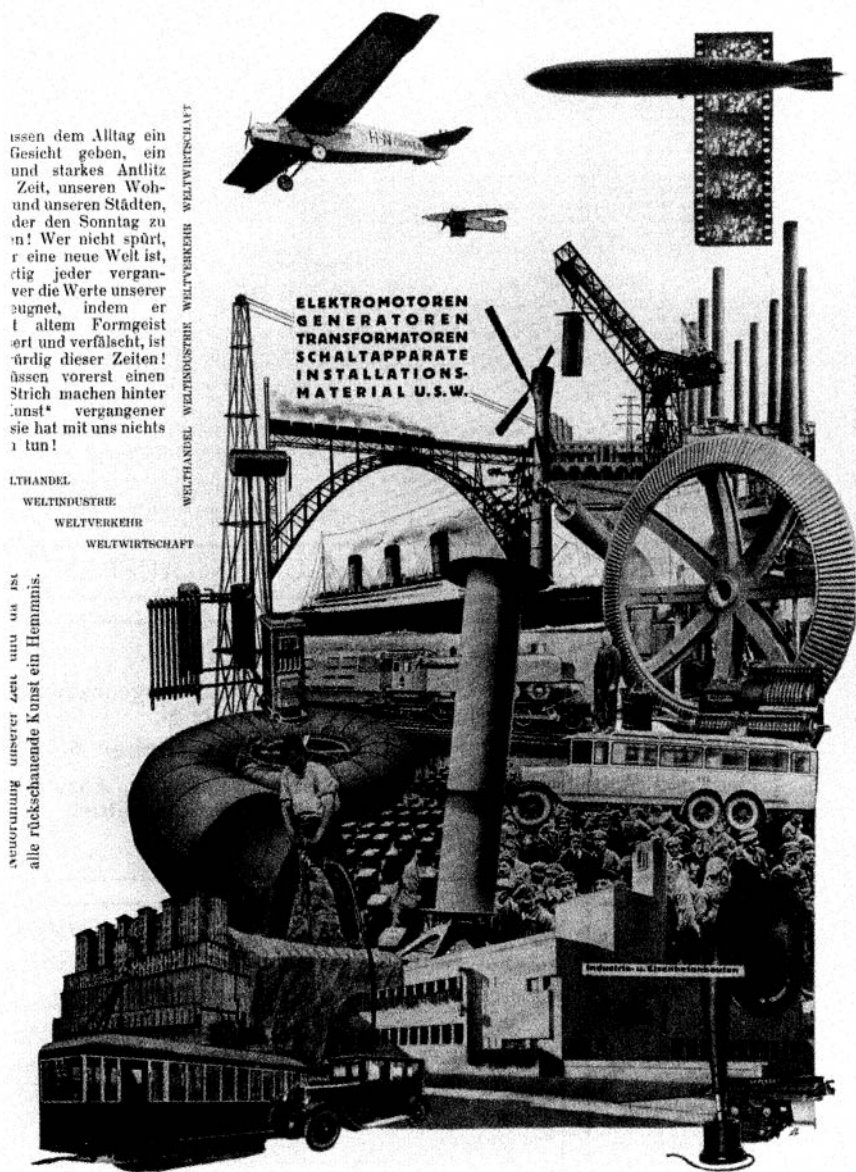


Figure 5. Detail from a modernist collage. From *Der Aufbau*: Österreichische Monatshefte für Siedlung und Städtebau, 1/1 (Vienna, February 1926).

persönlichen Lebens), it could only be through the thoroughgoing adoption of the technical. Those who wanted to “enter the promised land,” leaving the past behind, “will seize upon the formation of the new form of life [*Gestaltung des Lebens*] as a technical achievement. This is the thrust of the Bauhaus, unfettering the liveliest discussion and most vigorous efforts on all sides” (Neurath 1926a, 210–11).

In 1926, Neurath was thus deeply engaged in the definition and extension of a modernist world — at precisely the same time that he, Hans Hahn, and Philipp Frank joined Moritz Schlick in the formation of the “official” Vienna Circle. And it is this wider context of what one might call an Austrian and German “techno-modernism” that set the tone of the Vienna Circle “manifesto,” the preface to Carnap’s *Aufbau*, and much else in the work of the logical positivists of the mid-1920s.

The manifesto, “Wissenschaftliche Weltauffassung: Der Wiener Kreis,” which Neurath, Hahn, and Carnap drafted during the summer of 1929, deliberately politicized their philosophical stance when they argued that their position emerged from “fierce social and economic struggles.”¹⁰ They continued: “[O]ne group of combatants, holding fast to traditional social forms, cultivates traditional attitudes of metaphysics and theology whose content has long since been superseded; while the other group . . . faces modern times, rejects these views and takes its stand on the ground of empirical science.” Like so many of the architects writing in the various *Aufbau* journals, Neurath, Carnap, and their colleagues directly tied their enterprise to the “modern process of production, which is becoming ever more rigorously mechanised and leaves ever less room for metaphysical ideas” (Neurath 1973, 317). The echo of the *Aufbau* slogans of *Welthandel*, *Weltindustrie*, *Weltverkehr*, and *Weltwirtschaft* could not have been clearer. Or again, elsewhere in the same document: “We witness the spirit of the scientific world-conception penetrating in growing measure the forms of personal and public life, in education, upbringing, architecture, and the shaping of economic and social life according to rational principles” (ibid., 309). Rudolf Carnap shared some, but not all, of Neurath’s ambitions to construct the world anew. As the preface to *Der logische Aufbau der Welt* indicated, he held high hopes for that rather technical work, hopes that echoed all the left-technocratic cultural meanings of *Aufbau* I have discussed. It was, he argued, part and parcel of a rejection of the past and a “faith that the future belongs to this attitude”:

We do not deceive ourselves about the fact that movements in metaphysical philosophy and religion which are critical of such [a scientific] orientation have again become very influential of late.

Whence then our confidence that our call for clarity, for a science that is free from metaphysics, will be heard? It stems from the knowledge or, to put it somewhat more carefully, from the belief that these opposing powers belong to the past. We feel that there is an inner kinship between the attitude on which our philosophical work is founded and the intellectual attitude which presently manifests itself in entirely different walks of life; we feel this orientation in artistic movements, especially in architecture, and in movements which strive for meaningful forms [*Gestaltung des menschlichen Lebens*] of personal and collective life, of education and of external organization in general. We feel all around us the same basic orientation, the same style of thinking and doing. . . . Our work is carried on by the faith that this attitude will win the future. (Carnap 1967, xvii–xviii)

Imagine an American analytic philosopher of the 1950s resting his future reputation on the “inner kinship” between philosophy, collective life, and architecture. The thought-experiment fails. That Carnap’s *Aufbau* could be introduced this way is testimony to the enormous amount of cultural work that had already prepared the ground for these affiliations between fields to make sense — perhaps even to seem natural.

I do not, however, mean to imply that the postwar usage of *Aufbau* determined the content of the philosophical projects of that title. We know by now far too much of the much older empiricist and the more recent neo-Kantian roots of many aspects of Carnap’s project.¹¹ In the text of Carnap’s *Aufbau* itself, the author is, above all (as Michael Friedman has stressed), seeking to define a system in which each concept would have a fully specified relational and therefore objective place (Friedman 1987). By building up from basic elements, Carnap hoped to exclude the superfluous, the mystical, and indeed all that was not rationally or empirically located. It was this eliminativist strategy, this systematic construction from clear starting points, that girded Carnap’s *Aufbau*, broadly conceived, against reactionary assault.

As we now know, Carnap’s *Logische(r) Aufbau* originally had a very different title. The first draft manuscript of 1924 was labeled *Vom Chaos zur Wirklichkeit* and soon took on the more neo-Kantian appellation *Konstitutionstheorie* (see Coffa 1991, 208). By 1928, however, the term *Aufbau* would have carried a much wider connotation and attracted considerably more attention as it would quite clearly have suggested to German speakers, and to the Vienna Circle in particular, a set of associations that would have resonated across architecture, philosophy, and politics. Not unlike the term “postmodernism” in the 1980s, *Auf-*

bau in the mid-1920s embedded a work into a chain of meanings — and a wider cultural struggle (Huyssen 1986; Jones 1994). Having read the work, and very much approved of it, Schlick was reluctant to see it issued under its old moniker. On 4 January 1928, he strongly urged Carnap to change it:

As for the *title of the work*, I would like to say this: a book title not only has the task of correctly describing the work's content; it also must indicate the most important part of the content or the intent with suggestive force. This latter is the main thing. On these grounds I would decisively dissuade you from using the titles that you proposed at the close of your letter; they are too bland.... The fundamental thing in your book... [is] the general discussion of the principles of the constitution, to which the *Aufbau* of the system of knowledge is bound. For this foundation of principles, and more generally for the whole book, it seems to me that *Der logische Aufbau der Welt* is in fact the appropriate title. By these words it is, of course, to be understood that in the first instance the book has to do less with the actual and special development of a particular *Aufbau*, than with the principles of any such *Aufbau*.¹²

The idea of such a construction appealed to Schlick a great deal; as he had often stressed, the systematization of knowledge was just what was needed to root out bad reasoning, lies, and deception. As regards Schlick's suggestions, Carnap (never one to leap to a conclusion) hesitated, keeping his original title in the subtitle, but he eventually came to identify the work so closely with its new moniker that when the printers forgot to insert the *Konstitutionstheorie*, he didn't even notice. Increasingly, he had come to see this work in general, and the efforts of the Vienna Circle more generally, as part and parcel of a larger, critical cultural movement. Calling it *Der logische Aufbau der Welt* had come to be so natural, the old title had faded from view

2. *Aufbau* in America

People move across oceans with relative ease; complexes of ideas do not. When László Moholy-Nagy became director of the New Bauhaus in Chicago, he recruited Carnap as an occasional lecturer and brought on board in a continuing fashion three members of the Unity of Science movement, including Charles Morris.¹³ Throughout the 1930s, Morris had acted as an American clearinghouse for contact between Americans and the Vienna Circle, and by the time Carnap arrived in the United

States, Morris had become one of his closest philosophical contacts. In addition, for several years Morris had been the most active American in the ambitious series of conferences the Circle ran on the unity of science and was a coeditor (with Carnap and Neurath) of the successor journal to the Vienna Circle's *Erkenntnis: The International Encyclopedia for the Unity of Science*. Morris's philosophical program stressed the purification of language, the reduction of all talk to the simplest of starting points. Applying this very basic vision of the positivists, he tried his best to cast it as a variant of Dewey, presumably not to frighten his compatriots with a massive dose of philosophy with a German accent. In his courses at the New Bauhaus, he explicitly linked the goal of training in art to the reduction of language already imposed on science.

At the New Bauhaus and in his role as an editor of the *Encyclopedia of Unified Science*, Morris increasingly pushed his allies, especially Neurath, further from the political engagement that had marked the Circle's European incarnation. In a sense, the Americanization of the Unity of Science movement resulted from its now American funding and institutional setting, but the process was much more complex and interesting than that. In addition, Morris, Percy Bridgman, and other Americans infused pragmatic philosophy into the movement, while simultaneously the political pressure mounted to remove internationalism and Marxism from the movement. Partially a result of the generalized pressure to become scientifically "neutral," this depoliticization issued both from within the movement (Morris) and from without (Warren Weaver of the Rockefeller Foundation). By "depoliticization" I mean not just the deletion of Marxist or socialist ambitions, but a sharp narrowing of the broad cultural program of technological modernism that the left wing of the Vienna Circle had advocated. *Aufbau* had no significant role to play in the New Bauhaus.

Already in January 1935, when Morris first approached Weaver, he began to phrase the goals of the unified science program in terms that completely obscured these wider cultural ambitions: "I represent an international group of scientists and logicians interested in the unification of science[,] meaning by that term not a synthesis of factual results but the analysis and systematization of the language of science."¹⁴ Weaver encouraged Morris and supported the movement with more than words: by 1941, the movement had a twenty thousand dollar grant.¹⁵ Gone was the desire to meld a mode of art, architecture, and indeed a whole *Lebensgestaltung* into a language and demeanor. What was left was a tripartite program that included the coordination of scientists, the publication of an encyclopedia, and the establishment of a "unified center" that would stand against antisience.

With Carnap, Reichenbach, Frank, and Neurath on the move, Morris's established position in Chicago and his growing links with Rockefeller's Weaver and the University of Chicago Press gave him vastly increased power. This power was reflected in the tenor of his letters. Morris's correspondence with the senior philosophers in Europe had been extremely deferential. This changed. For, grateful as they were to be out of the Nazis' grasp, members of the Circle were precariously perched in their adopted countries. Even Carnap had a hard time securing a position and was in very difficult and sometimes acrimonious discussion about the terms of his employment with the University of Chicago. And he was probably the most secure of all the refugee positivists. To Morris, Carnap reported on the anti-Semitic situation at Princeton, one that effectively barred Reichenbach or Feigl from the appointment that Carnap had declined, leaving both Feigl and Reichenbach somewhat vulnerable.¹⁶

Through the placement of Carnap at Chicago, through the management of the other logical positivists, and through his own growing reputation, Morris's position in the movement changed drastically. Leaving behind his tentative approaches to the Circle in the late 1920s, he now began to instruct his collaborators on how they must act in the United States. For example, Morris resolutely resisted Neurath's attempt to push Reichenbach out of the new Unity of Science movement, on the grounds that parochialism had to be combatted: "I am interested in the Unity of Science movement as such, and not in any special group such as the Wiener Kreis."¹⁷

Elsewhere, Morris deprecated Neurath's proposed title for the encyclopedia, which began with the word "international." Some people (Morris did not specify who) did not like the very word "international," presumably because it smacked (in 1937) of unpatriotic leftism.¹⁸ Nor would Morris countenance Neurath's appellation of the contributors as "new encyclopedists" after the French *Encyclopédie*.¹⁹ Writing to Neurath, Morris reminded his Austrian colleague of his own earlier comment that the journal would proceed "without entering into discussions of principle touching political ideologies." Morris interpreted the meaning of this phrase as opening the *Encyclopedia* to all comers, with no particular political ideology to be associated with the journal — it was quite evidently a warning to Neurath that his political views, such as those expressed in *Der Kampf*, would have no place.²⁰ Indeed (according to Morris) if the journal needed anything, it needed *more science* and *more scientists* (not more politics and philosophy) in order to facilitate its acceptance in the wider scientific community.

Neurath's early pleas for a new *Encyclopédie* met resistance from

Morris, partly because Neurath understood and wanted to foster the revolutionary associations of its eighteenth-century counterpart. Voided of this association, the historical antecedent was much less threatening, as was evident from Morris's enthusiasm over a 1938 evaluation of the first volume of the *Encyclopedia* that appeared in the *New York Times Book Review*. The review (written by Neurath's relative W. Kaempffert) put it this way:

Indeed this work must serve a higher purpose than that of amalgamating the sciences for the benefit of specialists. The men who are making this encyclopedia have no desire to enter the political arena but every desire to influence intellectual leaders and to ensure freedom of scientific inquiry everywhere. The encyclopedists of the eighteenth century played their part in breaking the tyranny of their time without engaging in political struggles. So with this expression of empirical opinion.²¹

Here was a form of nonpolitical politics that Morris could endorse: tyranny would quake at the mere presence of scientific reasoning.

Neurath did not go gently into the Americanized, politically homogenized, and neutral Unity of Science movement. Repeatedly in his correspondence, Neurath returned to the old divisions between left and right, distinctions that for him still made a difference: "I remember in the WIENER KREIS as we read Wittgenstein, I again and again said: that is Metaphysics. It became dull and Hahn suggested I should speak of M. only to shorten the sounds[,] and since I too often said 'M,' he suggested I should only remark when I [was] satisfied by saying, 'NM.'"²² Tainted by Wittgenstein, others on the right side of the Circle also fell from Neurath's favor: Schlick's "Wittgensteinization and Decarnapization has been the real 'tragedy' of his later arguing. Wittgenstein now is, where I expected him to be, ... fighting scientific arguing and maintaining dull generalities on language, without any contact with empiricist science."²³

But if Wittgenstein's metaphysical proclivities tended to divide the left from the right wings of the Circle, Neurath envisioned a second order of boundaries between the Vienna Circle and sympathetic movements outside: "James, who is to a certain extent somebody who helps us, is on the other hand full of metaphysical trends, liking Bergson etc[.] [T]he same is right of Peirce. Therefore we should avoid to be in too close contact with a name ... which stresses the connection with these fine people, who nevertheless do not belong to our movement as such."²⁴

Outside the universe of "fine people" who might disagree about tac-

tics were the divisions that severed the Nazis from their enemies. Those philosophers who aligned themselves with the Nazi movement offended at an altogether different level:

It is a sad thing, that now the German illness enters the States [...] but that is the same in many other cases [i]n history etc. The Nazidom of Heidegger did not sufficiently shock people on the other side of the Atlantic. I bought his Nazi declarations, they are TERRIBLE, of the worst Goebbels type, full of distorted details, freely told lies etc[...] but you feel through all these sentences the HIGHER language of his philosophy.²⁵

As with so many leftists during the 1940s, Neurath's opposition to the right did not imply an easy alliance with the specific structure of the Soviet Union. Ever since the mid-1930s, it had been clear that, for Moscow, the Vienna Circle amounted to a variant form of *Machismus*, hardly a term of recommendation after Lenin's bitter denunciation of Mach in *Empiriocriticism*.²⁶ Neurath found his new political niche in a delicately perched third position — that is, as a technocratic left democrat. And here, he believed, the "Wissenschaftliche Weltauffassung" would ply its true trade: "I think our LOGICAL EMPIRICISM will play a certain role in the democratization on Europe."²⁷

The fourth and final division that Neurath drew during the war was, in some sense, against himself, against the earlier version of his own left wing. More precisely, politics had intervened. From a largely pro-Soviet Austromarxism, Neurath seems to have moved to a looser sort of decentralized and democratized socialism. His philosophy too turned away from even the hint of foundationalism that terms of reference seemed to suggest.

To a philosopher looking back today, it often appears self-evident that the project of an unmetaphysical philosophy was doomed to failure; there is, as Ernst Nagel has put it so well, no "view from nowhere." For the modern historian it seems equally apparent that the cry for an apolitical politics is similarly hopeless. Technocracy is not a day-pass from politics. One surviving visitor to the early Vienna Circle meetings (Peter Hempel) once told me that the Circle had an agreement to leave politics at the door. Our general view today is that politics, like the air we breathe, will not wait outside. And from art and design, we have also come to accept that there is no exemption from aesthetics. Where the Bauhäusler could stare into their lamps and chairs and see only pure function, we see an ideologically saturated aesthetic, frightening, inspiring, striking, even moving, grasping for an imaginary future. We cannot

any longer see the neutrality of pure function. Unaesthetic aesthetics, unpolitical politics, and unmetaphysical philosophy all seem elusive.

But taken together the various moments of modernism (captured by the call for an *Aufbau*) do convey a vision. It was a brief and fleeting vision, one held under fire. And that vision was given ever more coherence as the left wings of the Bauhaus and the Vienna Circle faced an increasingly hostile, murderously determined world.

The American welcome, while warm and for many lifesaving, was immensely complex and transformative. Pragmatists may have been sympathetic to positivists, but that did not rule positivism to be a form of pragmatism. The politics of a cautious ecumenical liberalism altered as it received. Whether challenging the “international” in the *International Encyclopedia of Unified Science* or questioning the right interpretation of “encycopedists,” Morris and his American colleagues subtly shifted Neurath’s modernist technocratic Austromarxism to a technical tool kit. And in others’ hands, the new *Lebensgestaltung*, dreamt of as a modernist *Aufbau*, slowly reduced to an academic specialty within the philosophy of science. The ideal of an *Aufbau*, as understood in mid-1920s Vienna, was conjointly architectural, political, and philosophical; it did not as such move across the Atlantic.

3. Summary and Conclusion

The problem is this: for any thesis about the association of two arenas of culture — whether it is telegraphy and electrodynamic theory, pneumatics and parliamentary politics, Weimar antirationalism and quantum uncertainty, thermonuclear weapons and particle physics, or architecture and philosophy — the question arises as to the status of cultural links.²⁸ One possibility would be that a claim that two groups are part of the same enterprise would amount to an assertion that there exists a relation of *logical* implication. For example, to claim that a Cartesian must oppose the intensely hierarchical Aristotelian cosmos follows from the basic Cartesian cosmological assumption that the universe is homogeneous. This does not seem appropriate to the philosophy-architecture links described here. Nothing in the tenets of logical positivism *implies* subscription to the basic assumptions of Bauhaus architecture.

Another possibility would be that one movement *caused* the other to take the form it did: either the Vienna Circle’s form of constructivism caused that of the Bauhaus, or the constructivism of the Bauhaus engendered that of the Vienna Circle. This too seems overly confining. Clearly there were logical positivists who had little interest in architecture one

way or the other. Worse yet — for a linear causal account of the link between architecture and logical positivism — the rallying cry to create an *Aufbau* became a slogan of the fascists in the mid-1930s, which would hardly suggest a unique bond between the two movements.

A far weaker assumption abdicates any form of explanation and simply identifies the two arenas as having developed in parallel. Unlike the logical or unicausal accounts, this structuralist one seems too weak; it loses touch entirely with the *work* that went into the establishment of relations between the two groups, their mutual legitimation, their common causes, and their enemies.

One possible alternative to the extremes of logical implication, uncausal determinism, and parallelism is to develop a notion of cultural meaning. That is, in the German-speaking world during the 1920s, a vocabulary and reservoir of images evolved that were shared by both the positivists and the Bauhäusler. This repository included a notion of transparent construction in which each element had its place and function. It included the aspiration of both groups to harmonize their own discipline (and interior life more generally) with the rationalism each saw embodied in what they perceived as the regularity, intelligibility, and functionality of contemporary technology. It included an emphasis on the role of the collective action of peoples. But these meanings and associations of meanings had to be *put* together and *held* together — there was nothing inalienably “rational” about a hydroelectric dam. The dam could have been (and was) deployed to signify the power of the unified nation under fascism, the productive energy of communism under Stalin, and the regenerative strength of America during the Works Progress Administration.

The point of examining these images of technology as *meanings* is twofold. On the one hand, meanings are not entirely aleatory. Groups cannot arbitrarily choose to make the icons of a society signify what they will. On the other hand, meanings are not fixed forever. The cultural meaning of *Aufbau*, Bauhaus architecture, and logical positivism all shifted radically — often to their practitioners’ horror — as the ideas moved outside the community in which they were forged.

To pursue a notion of cultural meaning then becomes at once a conceptual and historical task. It involves the articulation of the chains of associations that terms carried in a specific cultural location. *Aufbau*, to return to our example, not only maintains the age-old notion of building; it came, through the socialist city-state of Red Vienna in the post-World War I period, to include much more than mere bricks. An *Aufbau* reformed life, restructured values away from the decorative and superfluous and toward the practical; *Aufbau* in 1917–26 incorporated

rationalism and in general reached all the way down to the very form of life: *Lebensgestaltung*, *Lebensordnung*, and *Gestaltung der Lebens*. It was, as many authors emphasized, more than a revolution.

The historical-philosophical task of establishing the link between two arenas of culture (here logical positivism and modernist architecture) then becomes one of tracking the specific modes by which the two groups came to *share* cultural meanings: in this instance, through Neurath's work on *Aufbau* with leading city planners and architects; through Josef Frank's lectures and collaboration with Neurath; through Carnap's, Reichenbach's, and Neurath's visits to the Dessau Bauhaus; through the mutual political support offered between the logical positivists and Meyer's Bauhaus. If I could borrow a metaphor from chemistry, I would say that the Vienna Circle and the Dessau Bauhaus shared meanings the way two atoms share valence electrons. The philosophical-cultural-historical task of establishing a link becomes one of articulating the conditions under which concepts like *Aufbau* fit simultaneously into the meaning valences of both philosophy and city building. This relationship is, as advertised, not one of unilateral causation, logical implication, or deracinated *Zeitgeist*.

Indeed, without the experienced context of a postwar Vienna, a collapsed empire with its now illusory signs of aristocratic power, without the explosive building boom of Red Vienna and the ideological program that accompanied it, Americans did not and could not see the notion of *Aufbau* quite the way post-World War I Vienna did. It is not, I would argue, a mere accident that Rolf George translated Carnap's magnum opus as *Logical Structure*, not *Logical Construction*. Red Vienna had vanished, and the New World had its own fascination with universal, ahistorical, and apolitical *structures*.

In a different context, the great architectural historian Nicholas Pevsner once used the phrase "architecture of reason" to capture the faith many builders had that they could embody rationality in the very stuff of an edifice (Pevsner 1975). In a sense, the architects of the Bauhaus and Red Vienna aspired to a world that was the mirror image of the one sought by the Vienna Circle. Where the architects wanted to make buildings proceed like reason itself, excluding the decorative and dysfunctional, the philosophers of the Vienna Circle espoused spare, cool, and rational modern architecture as an emblem of their systematic construction of concepts, a reason of architecture. The *Aufbau*, the purifying moment of construction, stood for reformation in the domain that the Bauhaus wanted to substitute for art and in the field that the Vienna Circle hoped would replace philosophy.

In the first instance, the particular affiliation of the Vienna Circle and

the Dessau Bauhaus depended on specific cultural complexes in Austria and Germany just after World War I. But, more than that, their *continued* association was tightly bound to the cultural location of both groups and the efforts they made to maintain the sphere of meanings associated with the new unaesthetic aesthetics and unphilosophical philosophy — and it was here that terms like *Aufbau* functioned both as symbol and cement of that association. But when the external environment changed radically, through Nazism and emigration, the cultural meaning of the two movements shifted and constricted. After a brief moment together in the New Bauhaus in 1937–38, the two groups ceased to coexist in the same cultural sphere of early twentieth-century modernism. An international style of building in the early 1960s carried with it precious little of the resonance that Hannes Meyer might have wanted. And philosophy of science in the 1960s no longer self-consciously sought to participate in the reformation of the totality of culture, politics, and reason; nor did it seek to induce a reformation of daily life. Once in each other's valence of meanings, modern architecture and logical positivism had parted ways.

Notes

This chapter is an expanded version of P. Galison, "The Cultural Meaning of Aufbau," in F. Stadler, ed., *Scientific Philosophy: Origins and Developments* (Dordrecht: Kluwer, 1993). For helpful comments and suggestions, I would like to thank Caroline A. Jones. Research Assistants Richard Beyler, Cathy Carson, and Nani Clow were a great help. Jean Titilah assisted with the preparation of the manuscript. My thanks go as well to the Pittsburgh Archives for Scientific Philosophy (PASP), especially to Gerald Heverly. I am much indebted to the archivists at the University of Chicago (USMP designates the Unity of Science Movement Papers, Department of Special Collections, the University of Chicago Library; and PEP, the Peirce Edition Project, Indiana University–Purdue University at Indianapolis, which kindly provided access to many documents and letters relevant to Morris and the Vienna Circle. Material from the Carnap Collection of the Archives for Scientific Philosophy [ASP] is quoted by permission of the University of Pittsburgh.) Support for this research came from the Andrew W. Mellon Foundation.

1. One finds, for example, the less than immortal verses:

Aufsteht der Deutsche.
Befreit von der Fessel des Fröners,
reckt er die sehnigen Arme.
Angreift er sein Werk
nach seinem Gesetz.
Lasten bewegt er,
die schienen zu spotten
jedweder Kraft.
Berge versetzt er,
die wehrten den Weg ihm.
Und Spöttern zum Hohn,

zum Staunen der Welt,
wirkt er gewaltiges Werk.
(*Aufbau und Erbe* 1938, 6)

2. On *Siedlungen* and the new architecture, see Lane 1968, 58, 84ff.
3. For an excellent discussion of Neurath's involvement in the Bavarian revolutionary government and the links between Neurath's philosophy and his politics, see Cat, Chang, and Cartwright 1996.
4. On Neurath's involvement in the housing movement, see Hoffmann 1982, 140–48.
5. For details of the interactions between the Vienna Circle and the Bauhaus, see Galison 1990, 709–52; see also Galison 1986, 271–88; P. Neurath 1982, 289–97; and Stadler 1982b.
6. For more on Schuster, Schacherl, and Tessenow, see Hautmann and Hautmann 1980.
7. Wingler 1978, 3. Also on the masthead of the Viennese *Der Aufbau* was Generaldirektor Dr. Ing. Hermann Neubacher (Vienna).
8. This discussion draws from Marcuse 1986, 558–85.
9. There is an increasingly large and impressive literature on Neurath and his political engagement. See, for example, the following important works: Hegselmann 1983, 67–74; Hegselmann 1979; and Nemeth 1981.
10. For more on the Vienna Circle's implicit and explicit politics, see Wartofsky 1982.
11. On the relation of Carnap to Kant, see, for example, Coffa 1991; Friedman 1987, 521–45; Proust 1986; Haack 1977; Moulines 1985; Richardson 1992b.
12. Schlick to Carnap, 4 January 1928, Carnap Collection, ASP 029–30–36.
13. Engelbrecht 1973, 287. For more on the establishment of the New Bauhaus, see also Jordy 1969; see also *50 Jahre New Bauhaus*, Bauhaus Archiv, Argon: Berlin, n.d.
14. Morris to Weaver, 28 January 1935, box 1, folder 15, USMP.
15. Morris to Neurath, 29 September 1941, box 2, folder 1, USMP.
16. Carnap to Morris, 29 April 1936, PEP.
17. Morris to Neurath, 16 November 1936, box 1, folder 15, USMP.
18. Morris to Neurath, 17 April 1937, box 1, folder 16, USMP.
19. Morris to Neurath, 13 July 1937, box 1, folder 16, USMP.
20. Morris to Neurath, 26 August 1937, box 1, folder 16, USMP.
21. Quoted in Morris to Professor Robert C. Woellner, 22 October 1938, box 1, folder 17, USMP.
22. Neurath to Morris, 18 November 1944, PEP.
23. Neurath to Feigl, 26 February 1943, PEP.
24. Neurath to Morris, 18 November 1944, PEP.
25. Neurath to Morris, 18 November 1944, PEP; bracketed ellipsis is in original.
26. Neurath to Morris, 1 October 1934, box 2, folder 6, USMP.
27. Neurath to Morris, 30 May 1942, PEP.
28. Obviously, I choose these examples nonrandomly and will reflect on the work of Smith and Wise 1989; Shapin and S. Schaffer 1985; Schorske 1981; Forman 1971; and others.